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Maulana Azad National Institute of Technology, Bhopal

Department of Electrical Engineering

B.Tech. (2nd Year)

Mid-Term Examination, October 2024

Measurement and Instrumentation (EE-214)

Max. Marks: 40

Timing: 1.30 Hours

Q.1 (a) Define limiting error. State and explain the static characteristics of an instrument. (5)

(b) Explain elements of a generalized measurement system. Current was measured during a test as 30.4 A, flowing in a resistor of 0.105 Ohm. It was discovered later that the ammeter reading was low by 1.2 percent and the marked resistance was high by 0.3 percent. Find the true power as a percentage of the power that was originally calculated. (5)

Q.2 Describe the construction, operating principle and characteristic of linear variable displacement transducer. (10)

Q.3 (a) Briefly explain the method used to measure the depth of Ocean. (5)

(b) Describe the method used to measure the level of petrol in tank. (5)

Q.4 (a) Draw the phasor diagram of a current transformer. Drive the expressions for ratio and phase angle errors. (5)

(b) A 1000/5A ~~5A~~, 50 Hz current transformer has a bar primary and a rated burden of 12.5VA. The secondary winding has 196 turns and a leakage inductance of 0.96mH. With a purely resistive burden at rated full load, the magnetisation mmf is 16 A and the core loss excitation requires 12A. Find the ratio & phase angle errors. (5)